MATTHEW UTTI

TABLE OF CONTENTS

Collaboration quality …… 3

Justifying the collaboration quality indicators ……4

Reflecting on findings and future directions …… 6

Reference …….. 7

Measuring Collaboration Quality Between Students and Large Language Models

Conceptualisation, Reflection, and Analysis

This project follows a Design Science Research approach to evaluate a computational framework for measuring collaboration quality in interactions between students and Large Language Models. DSR is appropriate as the goal is to design a set of measurable indicators that enable collaboration quality within learner-LLM dialogue.

Collaboration Quality

High-quality collaboration between a student and a large language model in a learning task involves an iterative dialogue that encourages layered thinking, gradual development, and incremental refinement of ideas. It's a practice that frowns at transactional use cases where a student submits a single prompt and accepts the response without an in-depth thought process towards shaping the solution.

Large Language Models are increasingly being integrated, both actively and passively, into higher education learning environments, offering students opportunities for on-demand problem-solving support. However, the effectiveness of these tools depends largely on how students interact with them. Emerging research suggests that productive learning with LLMs occurs when students engage with them as collaborative thinking partners rather than passive answer providers. In this context, collaboration can be understood as a process in which a learner interacts with an external entity towards a shared educational objective (Oppenheimer, D.M, 2025).

A study examining undergraduate students interacting with AI assistants in a software engineering course found that students used LLMs extensively during early stages of development tasks, with usage declining across project milestones. This indicates that collaboration patterns evolve over time and that simple usage frequency cannot fully capture collaboration quality. The same work also observed that AI-generated solutions tended to exhibit lower computational complexity, suggesting that reliance on AI outputs may influence the depth of student problem-solving strategies. These findings highlight the importance of examining not only whether students use LLMs but also how they integrate them into their reasoning processes (Naman, A., 2025).

An analysis of LLM effects on students across qualification, socialization, and subjectification stressed that engaging in activities such as self-explanation, iterative refinement, and recall practices promotes deeper cognitive processing and long-term retention. In other words, passive acceptance of AI-generated responses may lead to superficial understanding and short-term performance gains that do not translate into durable learning outcomes (Huang et al., 2025).

Justifying the Collaboration Quality Indicators

To deconstruct what meaningful collaboration between learners and the LLM looks like in terms of how the dialogue evolved, who drove the conversation, and whether collaboration actually deepened over time, the following analysis focuses on four key indicators:

- Student by engagement level
- Response adequacy by engagement
- Request progression by engagement
- Mean word count by engagement

The choice of indicators is rooted in patterns observed during the exploratory data analysis. Early visualisations not only revealed when students used the LLM and how often they interacted with it, but also captured the quality of collaboration unfolding within those interactions.

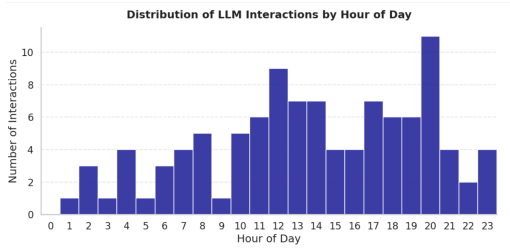

Fig 1.0

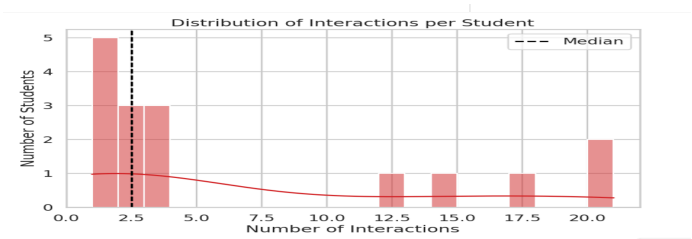

Fig 2.0

The histogram of interactions by hour of day (Fig. 1 .0) showed clear peaks around midday and again in the evening. These time windows show visible activities between 10 AM and 2 PM, and between 5 PM and 8 PM, corresponding to periods when students are likely actively working on academic tasks such as outlining essays and preparing for assessments.

However, the histogram analysis on the distribution of interactions per student (Fig. 2 .0) lent more insights - While some students returned repeatedly, the majority interacted only once or a few times. Most users clustered around 1–3 interactions, with the median near 3–4, indicating that many students used the LLM sparingly. However, only a minority engaged in 17 or more interactions. This contrast suggests two distinct behaviours: transactional use, where students ask once and leave, and iterative collaboration, where students repeatedly refine ideas.

Hence, if collaboration is concentrated at certain times of day, and if only some students engage repeatedly, then the next step should examine what actually happens inside those interactions.

Students by Engagement Level: Answers the question - Who Actually Collaborates?

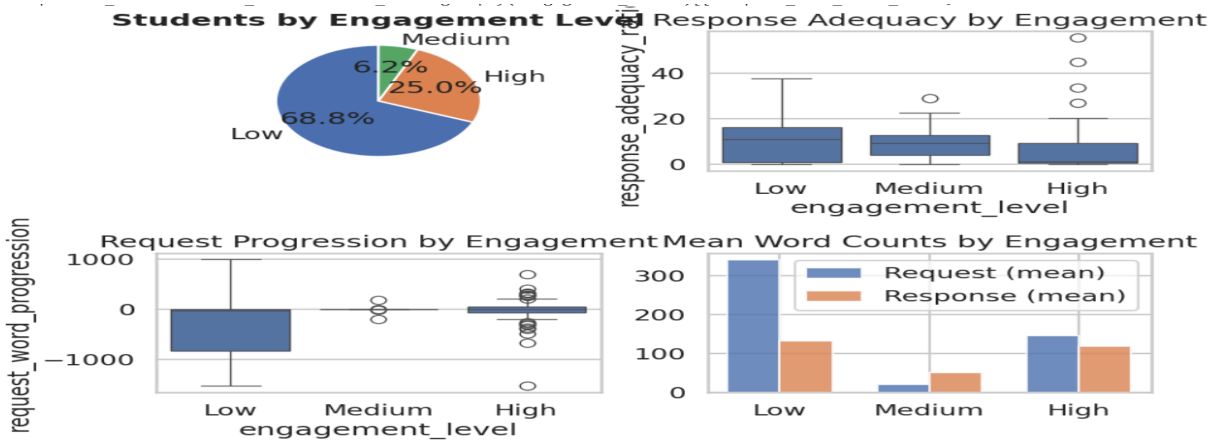

Fig 3.0

From Fig. 3 .0, the distribution of students by engagement level shows that most students fall into the low-engagement group, with a much smaller proportion appearing in the medium and high engagement categories. This is an important starting point because collaboration is inherently iterative. A single prompt and response may be helpful, but it does not suffice for what can be perceived as a true dialogue. The dominance of low-engagement users suggests that many students approach the LLM transactionally: they ask one question, receive an answer, and move on. In contrast, the high-engagement group, though smaller, represents students who return repeatedly, while refining their ideas and engaging in back-and-forth interactions. This distinction matters because collaboration quality is not only about what is said, but also about whether ideas are co-developed over time.

Response Adequacy by Engagement: Answers the question - Who Is Doing the Work?

Once engagement is established, the next question becomes: who is contributing more: the learner or the LLM? The response adequacy plot on the top-right corner of Fig. 3 .0 addresses this by comparing the length of LLM responses relative to student prompts.
The variability is striking within the realm of low-engagement interactions as it reinforces the idea that some students provide short prompts and receive long, detailed responses. Hence, LLM often compensates for limited student input. While helpful, this dynamic is less collaborative and more assistive, as the learner contributes minimally to the development of ideas.
In contrast, high-engagement students display more stable response adequacy ratios. Their interactions appear more balanced, indicating that the conversation evolves through mutual contribution. Ultimately, the learner refines prompts, and the LLM responds accordingly in a more dialogic structure.

Request Progression by Engagement: Answer the question - How does the Dialogue Evolve?

Analysis from the bottom left corner of Fig 3.0 shifts focus from balance to development over time. If collaboration is meaningful, student prompts should change as ideas become clearer.

The visualisation reveals that Low-engagement students show some negative progression, indicating little continuity between prompts with accompanying minimal conceptual build-up. On the flip side, highly engaged students demonstrate more positive progression as their prompts grow in complexity. This pattern reflects learners who are actively refining their thinking, incorporating previous responses, and expanding their arguments. In other words, the conversation is not static - it rather evolves.

Mean Word Counts by Engagement:
The plot at the bottom right corner of Fig. 3.0 compares the mean word counts and provides an insightful contrast between engagement groups. Low-engagement students tend to submit longer initial prompts but engage less frequently. This reinforces the fact that they paste a large amount of text once and rely on the LLM for a single response. High-engagement students, on the other hand, tend to use shorter prompts but return repeatedly. Their interaction style shows a balance of interactive conversation rather than a one-off request.

Reflection on Findings and Future Directions

The analysis revealed a distinction between transactional use and iterative collaboration. Most students interacted with the LLM only sparingly for one-off information retrieval. In contrast, a smaller group engaged in repeated interactions, using shorter prompts that evolved into full-scale collaboration. A key surprise was that longer prompts did not necessarily signal better collaboration; instead, meaningful engagement appeared to depend on iterative dialogue rather than single, detailed queries. However, ambiguity remains because structural metrics like word counts and interaction frequency do not fully capture conceptual depth. To improve the analysis, future work should incorporate semantic evaluation of prompts and responses to track how ideas develop across each interactive turn rather than relying solely on length-based indicators. Linking interaction patterns to learning outcomes like essay quality or grades would also strengthen conclusions about collaboration effectiveness. Also, analysing session-based time-series behaviour around deadlines could reveal how collaboration evolves during learning cycles. Evaluating collaboration quality at scale would require larger datasets, full conversation threads, assignment context, and student performance data. These would future analyses beyond measuring engagement to assessing how Learner–LLM collaboration genuinely supports deeper understanding.

REFERENCE:

The corrected Harvard reference is: Oppenheimer, D.M., Cash, T.N. and Connell Pensky, A.E. (2025) 'You've got AI friend in me: LLMs as collaborative learning partners', International Journal of Artificial Intelligence in Education, 35(6), pp. 3896–3921. Available at: doi.org.

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